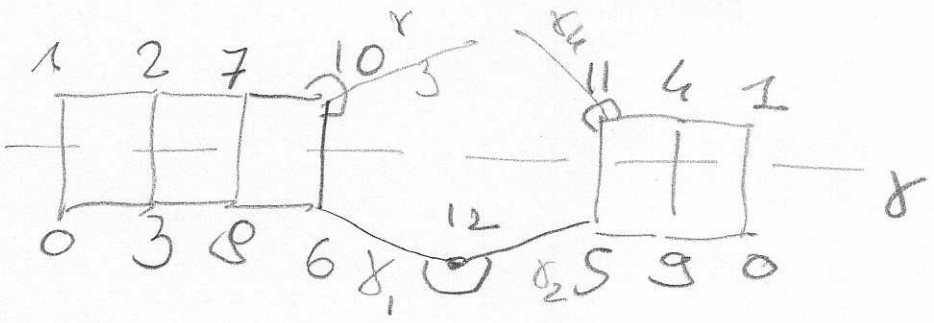
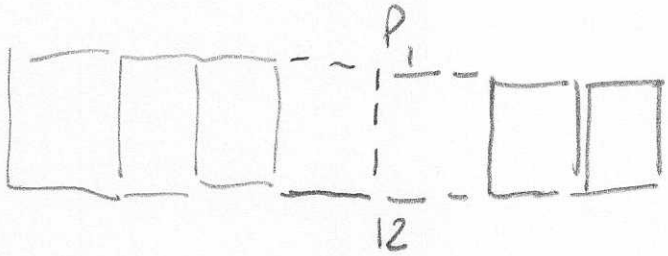
Diagram 3219



We consider the equator γ passing through the midpoints of the segments 10, 49, 115, 106, 78, and 23

A mirror symmetry with respect to γ maps the equator γ_1, containing the edge $6'12'$ into the equator γ_3 and the equator γ_2 into γ_4. In particular γ_3 and γ_4 intersect transversally in the two pts P_1 and P_2 and 12 is mapped either in P_1 or in P_2. Note that 12 must be different from both. Assume $s(12) = P_1$; then there is a Hyle equator γ passing through P_1 and 12

Since P_1 and 12 have the same longitude with respect to γ we conclude that γ is orthogonal to δ. Note that the bisector of 12 contains both p_{12}. So it must be γ. In particular this means also that '10' & P_1, '12' is a quadrilateral congruent to the others, as it is '11' P_1 5' 12':



But each quadrilateral has area $\frac{2\pi}{3}$, so a sphere cannot contain 7 of them